SURYA TEJA TIRUVEEDHULA

Software Engineer | suryatejatiruveedhula03@gmail.com |+91 9392317010 | linkedin.com/in/suryatejatiruveedhula-3a04901bb/

TECHNICAL SKILLS

Programming Languages: C#, C++, Python, JavaScript (JS), HTML

Frameworks & GUI Development: .NET, WPF, XAML, ASP.NET, Blazor, Qt/QML, Flask, MVVM

Database Management: SQL, Microsoft SQL Server (MSSQL), PostgreSQL, pgAdmin

Communication & Instrument Control: TCP/IP, SCPI, VISA, LAN-based Instrument Communication, DDC Bus Communication, MIL-STD-1553B

Data Visualization & Analysis: OxyPlot, Tableau, MATLAB, PDW Data Visualization, TDOA/DF Analysis, RFTMS Data Analysis

Development Tools & Version Control: Microsoft Visual Studio, Visual Studio Code (VS Code), Git, Wireshark

RF, ATE & Engineering Software: MESG, Pulse Building, Keysight Expert, SIMVIEW Z9500, IQ Tools, STK (Systems Tool Kit), Cisco Packet Tracer

System Integration & Testing: Automated Test Equipment (ATE), Equipment Controller Development, Hardware–Software Integration, Telemetry Data Acquisition, Functional Testing, System Verification, Technical Documentation

CLIENTS WORKED WITH

DLRL, ELSEC, RCI, BRAHMOS, SDI, LRDE, HAL, BEL, SAC, DEAL

PROFESSIONAL EXPERIENCE

Pinaka Aerospace | Software Engineer — Nov 2025 – Present

Hyderabad, India

1. SDI, Indian Air Force – Software Defined Radio (SDR) ATE

- Developed GUI functionalities for SDR-based ATE, supporting radio configuration, communication interfaces, and automated test-case execution.
- Supported requirement analysis, functional verification, debugging, and test documentation SDR,SRS,UML,SDD,PDR for Air Force-specific testing requirements.

2. DLRL, DRDO – ELSEC SAMOOHA: 4-Port Coherent Signal Generator

- Developed and integrated software functionalities for multi-channel coherent signal generation, configurable time delays, and time/phase/amplitude calibration for RF/EW simulation and payload testing.
- Implemented instrument-control workflows and emitter/scenario configuration; contributed to SRS/SDD/UML documentation, calibration procedures, integration testing, and DRDO technical support.
- Authorized Personal to Use ITAR Keysight with Proper Access.

3. DEAL, DRDO – RF Test and Measurement Systems

- Contributed to RF test-system software, instrument communication, measurement workflows, and system integration activities as per project requirements.
- Prepared and reviewed technical documentation SRS/SOP/SDD, software requirements, command descriptions, and test procedures to support implementation, verification, and project deliverables.

4. BrahMos – Telemetry Data Acquisition and PDW Telemetry Station

- Developed and integrated telemetry acquisition functionalities for Ka-band and Ku-band PDW telemetry applications, supporting data collection and analysis for DF/TDOA evaluation.
- Coordinated system communication and integration with Navy stakeholders; supported interface verification, troubleshooting, and technical documentation.

5. HAL – Su-30MKI IRST / IDMG UUT Test System

- Analysed system requirements and contributed to GUI and maintenance-functionality modifications for IRST/IDMG UUT testing, including MFD interfaces and MIL-STD-1553B, ARINC 818, and STANAG-related requirements.

- Prepared technical proposals, interface clarifications, and pre-bid documentation covering test-system architecture, communication interfaces, and integration responsibilities.

6. DLRL, ELSEC – Radome Controller and Seeker Acquisition Testing

- Developed a radome-controller GUI with Custom and Radiation modes; supported servo-controlled testing, VNA trace acquisition, and phase measurement analysis across angular positions and TX/RX configurations.
- Supported RF test-system integration, measurement verification, troubleshooting, and preparation of test procedures and technical documentation.

7. Technical Proposals and Customer Coordination

- Collaborated with engineering and marketing teams as a technical representative for defence-sector pre-bid meetings, requirement analysis, and solution discussions with DRDO and HAL.
- Prepared technical proposals, compliance inputs, scope clarifications, and Bill of Quantities (BOQ) inputs to support tender submissions and project acquisition.

Astra Microwave Products Ltd | Software Engineer — Jun 2024 – Nov 2025

Hyderabad, India

1. Checkout System – Equipment Controller, DDC Communication and PDW/TDOA Analysis

- Developed Equipment Controller (EC) GUI modules to configure, control, and monitor 11+ equipment units and associated subsystem modules through integrated software interfaces.
- Implemented DDC card integration and bus-communication workflows to acquire raw data packets from DR systems, with TLK View used for packet monitoring and data inspection.
- Integrated MESG software APIs for equipment control, parameter configuration, subsystem communication, and payload-related test operations.
- Developed SQL database functionality for user login, authentication, operational logs, equipment status records, and test-data traceability.
- Supported DR testing, acquired-data processing, and visualization of Pulse Descriptor Word (PDW) parameters, enabling analysis of Time Difference of Arrival (TDOA) and Direction Finding (DF) performance.
- Configured payload test parameters and supported emitter placement scenarios involving 500+ emitters distributed across multiple locations in India, incorporating geospatial positioning and satellite visibility considerations.
- Performed calculations and scenario analysis for Low Earth Orbit (LEO) satellites and geostationary (GEO) reference satellites to support emitter visibility, coverage assessment, and PDW/TDOA evaluation.
- Contributed to subsystem integration, functional verification, debugging, test documentation, and end-to-end checkout workflows spanning equipment control, data acquisition, and visualization.

2. ANVESHA - SSPA Control and RF System Integration

- Developed software functionalities for SSPA control, equipment configuration, RF test operations, and integration with signal-generation and measurement systems.
- Supported equipment communication, system testing, troubleshooting, and technical documentation for RF test and measurement workflows.

Advantum Health | Data Analyst — Jan 2024 – Jun 2024

Hyderabad, India

- Developed automated workflows for managing provider profiles in CAQH databases, ensuring credentialing compliance.
- Built backend systems to validate DEA and CDS licenses, reducing manual review time and improving data reliability.
- Credentialing standards, resulting in faster provider onboarding cycles and reduced rejection rates.
- Managed Website for Providers US Healthcare – DB's.

EDUCATION

Vellore Institute of Technology (VIT) — **Sep 2020 – Jul 2024**

Bachelor of Engineering in Computer Science and Engineering - AI **CGPA: 8.9**

10th: *Sri Chaitanya/CGPA:* ***9.8 (2017-18)***

Intermediate: *Ratnam JR College /CGPA:* ***9.5(2018-2020)***

CERTIFICATIONS

- ASP.Net for Engineers - Coursera. • Blockchain Developer - NASSCOM.
- Java Programming: Solving Problems with Software - Duke University.
- Tableau _ Smart Bridge for Data Visualization.
- C# WPF For beginners – Coursera
- RF Instrument signal and systems - Keysight conference
- Signal Waveform generator and Usage of API BY RnS.
- Data analytics & Visualization Using PostgreSQL - Udemy.

HONOURS & AWARDS

- **Token of Appreciation – DLRL, ELSEC:** Recognized for excellence in UI development for system control, in the presence of Tri-Services dignitaries.
- **Technical Project Contribution**: Contributed to defence and aerospace projects involving RF test automation, telemetry data acquisition, equipment control, and system integration.
- **Technical Proposal & Pre-Bid Support:** Contributed as a technical representative in defence-sector pre-bid activities, tender documentation, and customer requirement analysis
- Awarded Token of Appreciation at **DLRL ELSEC in front of Tri-Services dignitaries** for excellence in UI development for system control.
- Acknowledged for excellence in **secure system design and real-time payload management** used in national defense operations.
- **Smart India Hackathon 2023;** recognized for innovative problem-solving in a national-level competition.
- Smart Baby Cradle (VIT Innovation Clinic 2024); project selected from 600+ entries and acquired by a cradle company.
- Held leadership roles as Lead Coordinator for IEEE and IETE Chapters; Organized 63+ technical events and hackathons.